WEEK 1 · DAY 2 · SHIP AND OCEAN SYSTEMS DESIGN

# Plan of the Day

Friday 6 November · Sessions W1.3 and W1.4 · 13:30–17:00 · room VA3
Bring a laptop: the exercises are done in Python or Excel

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# The Plan of the Day

## W1.3 · 13:30–15:30 · VA3 · Process and sizing

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| 13:30 | **Lecture 1.3 · The design process**<br>From Wednesday: design spiral, phases A to D, contract plans |
| 13:50 | **Activity 1.3 · SFI Explorer, in pairs** |
| 14:00 | **Lecture 1.4 · Sizing from regressions**<br>Two routes; the Appendix A fits; a bulk carrier and a 2,000 TEU ship |
| 14:40 | **Break · 10 min** |
| 14:50 | **Exercise 1.4 · 25 min**<br>The reference ship from the tanker fits; try 5 min, then together |
| 15:15 | **Closing the gap by hand**<br>One factor k, then the design spiral |
| 15:25 | **Break · 5 min** |

## W1.4 · 15:30–17:00 · VA3 · The equation

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| 15:30 | **Lecture 1.5 · The design equation**<br>Buoyancy = weight; the weights as powers of Δ; iterate |
| 15:50 | **Exercise 1.5 · 50 min**<br>Review of the two methods, then Q1–Q5 alone (Q6 extended); answers together |
| 16:40 | **Project groups · final today**<br>Check your group on the course site |
| 16:45 | **Notebooks in Google Colab**<br>Where the notebook activities are on the site, and how to open them in Colab |
| 16:50 | **Close · after class and prepare for Wednesday on the site** |
| TBC | **Moved · Ship types 1.3 and 1.4 to Wednesday 11 or Friday 13 Nov, TBC** |